

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Given $y = f(u)$ and $u = g(x)$, find $\frac{dy}{dx} = f'(g(x))g'(x)$ for the following functions.

$$y = \sin u, u = 8x + 9$$

The chain rule states that if $f(u)$ is a differentiable function at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x and the derivative is shown below.

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

Write the chain rule in Leibniz's notation for $y = f(u)$ and $u = g(x)$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}, \text{ where } \frac{dy}{du} \text{ is evaluated at } u = g(x)$$

Given $y = \sin u$, find $\frac{dy}{du}$, the derivative of y with respect to u .

$$\frac{dy}{du} = \cos u$$

Find $\frac{du}{dx}$, the derivative of u with respect to x .

$$u = 8x + 9$$

$$\frac{du}{dx} = 8$$

Finally, use the chain rule to find $\frac{dy}{dx}$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= (\cos u) \cdot 8 \\ &= 8 \cos(8x + 9) \end{aligned} \quad \text{Replace } u \text{ with its expression.}$$

Therefore, $\frac{dy}{dx} = 8 \cos(8x + 9)$.